Soy Isoflavones and Risk of Cancer Recurrence in a Cohort of Breast Cancer Survivors: Life After Cancer Epidemiology (LACE) Study

Purpose Soy isoflavones, structurally similar to endogenous estrogens, may affect breast cancer through both hormonally-mediated and non-hormonally related mechanisms. Although the effects of soy are not well understood, some breast cancer survivors increase their soy intake post-diagnosis in attempt to improve their prognosis. Therefore, we examined the role of soy isoflavone intake and the risk of breast cancer recurrence by hormone receptor status, menopausal status, and tamoxifen therapy. Materials and methods A cohort of 1954 female breast cancer survivors, diagnosed during 1997–2000, was prospective followed for 6.31 years and 282 breast cancer recurrences were ascertained. Isoflavone intake was assessed by mailing modified Block and supplemental soy food frequency questionnaires to participants, on average 23 months post-diagnosis. Risk of breast cancer recurrence, measured by hazard ratios (HR) and 95% confidence intervals (CI), was estimated using multivariable delayed-entry Cox proportional hazards models. Results Suggestive trends for a reduced risk of cancer recurrence were observed with increasing quintiles of daidzein and glycetin intake compared to no intake among postmenopausal women (P for trend: P = .08 for daidzein, P = .06 for glycetin) and among tamoxifen users (P = .10 for daidzein, P = .05 for glycetin). Among postmenopausal women treated with tamoxifen, there was an approximately 60% reduction in breast cancer recurrence comparing the highest to the lowest daidzein intakes (>1453 micrograms (µg)/day versus < 7.7 µg/day) (HR, 0.48; 95% CI, 0.21–0.79, P = .008). Conclusion Soy isoflavones consumed at levels comparable to those in Asian populations may reduce the risk of cancer recurrence in women receiving tamoxifen therapy and moreover, appears not to interfere with tamoxifen efficacy. Further confirmation is required in other large prospective studies before recommendations regarding soy intake can be issued to breast cancer survivors.